Model Orthogonalization: Class Distance Hardening in Neural Networks for Better Security

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***Abstract*—The distance between two classes for a deep learning classifier can be measured by the level of difficulty in flipping all (or majority of) samples in a class to the other. The class distances of many pre-trained models in the wild are very small and do not align well with humans' intuition (e.g., classes turtle and bird have smaller distance than classes cat and dog), making the models vulnerable to backdoor attacks, which aim to cause misclassification by stamping a specific pattern to inputs. We propose a novel model hardening technique called model orthogonalization which is an add-on training step to pretrained models, including clean models, poisoned models, and adversarially trained models. It can substantially enlarge class distances with reasonable training cost and without much accuracy degradation. Our evaluation on 5 datasets with 22 model structures show that our technique can enlarge class distances by 177.63% on average with less than 1% accuracy loss, outperforming existing hardening techniques such as adversarial training, universal adversarial perturbation, and directly using generated backdoors. It reduces 80% false positives for a state-of-the-art backdoor scanner as the enlarged class distances allow the scanner to easily distinguish clean and poisoned models, and substantially outperforms three existing techniques in removing injected backdoors.**

I. INTRODUCTION

A backdoor in a deep learning model makes any inputs stamped with a specific pattern to be misclassified to a target class. While adversarial sample attack requires generating perturbations on the fly to cause an input sample, e.g., a video frame, to be misclassified, backdoor attack can have prompt effect by simply stamping a pattern. While backdoors can be injected through various methods, such as data poisoning [1]–[5], clean label poisoning [6]–[9], and neuron hijacking [10], they widely exist in naturally trained models (see Section II). We call them *natural triggers*. Natural triggers could be due to (1) the similarity between classes, e.g., a small fixed patch on any dog images can make the classifier predict cat, and (2) the model undesirably learning strong low-level features. We will show in Section II that there is a small backdoor between the turtle and bird classes even though they are unlike in humans' eyes. With the increasing applications of deep learning models in security-critical tasks such as autonomous driving [11], [12], surveillance [13], [14], access control [15], etc., backdoors are becoming a prominent security threat.

Existing defense techniques can be categorized to *backdoor scanning* that determines if a model has an injected backdoor [16]–[24], *backdoor attack detection* that determines on-the-fly if an input contains a backdoor pattern [24]–[35], and *backdoor elimination* that removes an injected backdoor [36]–[39]. Most these techniques focus on defending *injected* backdoors. For example, backdoor scanners Neural Cleanse (NC) [40] and ABS [41] rely on the assumption that injected backdoors tend to be small as they want to be stealthy. Backdoor attack detection techniques such as NIC [25] and SCAn [26] rely on the observation that an input sample stamped with a backdoor likely causes different model internal behaviors than a clean sample. Elimination techniques such as NAD [39] rely on benign samples to suppress injected backdoors. Detailed discussion of these techniques can be found in Section VII. However, they are much less effective for natural backdoors which are not injected but rather due to problems in training and even the nature of data. For example, natural backdoors cause a lot of false warnings for scanners as they cannot distinguish natural and injected backdoors [42]; a sample with a natural backdoor may likely evade detection as natural backdoors are usually benign features that may not induce abnormal internal behaviors; and using clean data cannot eliminate natural backdoors which are rooted exactly in the clean data.

Adversarial training is a widely used technique for model hardening which can force a model to unlearn unrobust (low-level) features. It aims to enforce any input undertaking adversarial perturbations in an L^p bound to be correctly classified by the hardened model. According to our studies (Section II and Section V), the improved robustness can help mitigate backdoors including natural backdoors. However, due to its L^p bounded training, which only considers local perturbations around individual samples, the protection against backdoors is limited. Also, it is known that adversarial training may cause non-trivial model performance degradation. In addition, our study shows that directly using backdoors generated by scanners to adversarially train a model does not work well due to either the extremely high computation cost or the longer convergence time. Intuitively, using a backdoor (which denotes much larger perturbations compared to those in adversarial samples) in adversarial training is like imposing a substantial displacement of the decision boundary. If not done properly, the decision boundary will oscillate. And since natural backdoors could exist in between any pair of classes, the training has quadratic complexity, which is very costly when the number of classes is large.

We propose a novel model hardening method to improve resilience to backdoors. It is an add-on training step for pre-trained models, including adversarially pre-trained models. It substantially suppresses backdoors, including both injected and natural backdoors, making backdoor attacks more difficult (e.g., having to use much larger patterns and/or easier to be detected by scanners), with reasonable training cost and without substantial accuracy degradation. Specifically, we consider the minimal backdoor between two classes (that flips samples in a *victim class* to a *target class*) the *distance between the classes*. As such, we aim to achieve the possible maximum distances for all class pairs. Intuitively, if we project all high dimensional data points to a 2-dimensional space, *a decision boundary allows the maximum distance between two classes when it is perpendicular to the line between the centers of mass of the two classes* (visualization and more explanation in Section III). We hence call the process of finding such a decision boundary *model orthogonalization*. Our technique hardens all class pairs for a model. For each pair a and b, it repetitively generates the minimal backdoors from a to b and from b to a and adversarially trains the model with the two backdoors. In other words, it forces the model to gradually unlearn the low-level backdoor features (and focus on high-level features with more semantics). The symmetric training of the two directions of a pair substantially alleviates oscillation and improve effectiveness. To address the high computation cost induced by the quadratic complexity, our technique features a scheduler that selects a most promising pair to harden in each training round based on the potential of distance enlargement. It leverages the observation that different class pairs have different distance capacities. For example, a pair of turtle and bird has more potential than a pair of cat and dog as the latter two are so close that their distance is hard to enlarge no matter how much training effort is spent. A number of methods are further used to speed up the training process such as reusing backdoors from previous rounds and dynamically adjusting bounds. Details are in Section IV.

Our contributions are summarized as follows.

- We intuitively and formally define the problem of model orthogonalization, which is an add-on training step. As part of it, we define the distance between a pair of classes, which serves as the basis for the hardening process.
- We devise a new training process to achieve orthogonalization. It features symmetric training (training the two directions of a pair together), pair scheduling, and a few other designs.
- We develop a prototype Moth (Model OrTHogonalization). Our evaluation on four standard datasets and six different model structures shows that Moth can improve class distance for naturally trained models by 119.87% and adversarially trained models by 52.87% with less than 1% accuracy degradation on average. With similar hardening performance, our technique is 9x faster than a baseline that does not use scheduling. It achieves 29.72% more distance improvement than a baseline that does not perform symmetric training. It can achieve 95.80% more distance improvement and 2.58x faster than using *universal adversarial perturbations* [43]. We also conduct experiments on 30 pre-trained models downloaded from the TrojAI competition [42], a competition for backdoor scanning. Moth improves the class distance by 232.39% over the original models and is 11x faster than the baseline. We apply Moth in two applications including reducing false positives for backdoor scanning and eliminating injected backdoors in existing models. It can reduce false positives by 81.25% for the first application. Regarding the second application, the attack success rate (ASR) of injected backdoors is reduced (by orthogonalization) from almost 100% to 1% on average, outperforming three state-of-the-art backdoor elimination approaches with the best performance of reducing ASR to 26.75% on average. Moth is publicly available at [44].

Threat Model. We consider backdoors between individual pairs. That is, a backdoor can flip samples from a victim class to a target class, called *label-specific backdoor* in the literature [40], [41], [45], which is more general than *universal backdoor* that flips any sample of any class to a target class. We consider backdoors that are either injected (in a poisoned/hijacked model) or naturally present (in a clean model). We consider both are equally harmful. Our goal is to enlarge class distances such that it is more difficult to find backdoors, without sacrificing much accuracy. That is, to launch attack, the attacker needs to use a large pixel pattern that may already possess a lot of semantic features of the target class.

In this paper, we only consider static backdoors, in which the backdoor patterns are input agnostic, like patch backdoors [1]. There are dynamic backdoors such as reflection backdoors [3], composite backdoors [4], and feature space backdoors [46]. We conduct a preliminary study on a few dynamic backdoors and Moth can reduce the ASR to some extent (see Section VIII). We will leave more exploration to our future work. We argue that our contributions are still very valuable because model hardening for static backdoors is still an open problem. Finally, we assume only a subset of the original training dataset (5%) is available when Moth is used to remove injected backdoors.

II. MOTIVATION

Figure 1 shows sample images from a normally trained ImageNet model downloaded from a widely-used model repository [47] and its natural backdoors (derived using NC [40]). The first column shows the backdoors, with the first row flipping dog images to cat, the second one turtle to bird, and the third one cat to bird. The second and third columns show the victim class samples; the fourth and fifth columns the victim class samples stamped with the backdoor patterns; the sixth column the target class samples, and the last column the size of backdoor in terms of the aggregated pixels in the R, G and B channels (i.e., L^1 norm). For instance, a trigger of size 615 (second row) has around $615/3 = 205 \approx 14 \times 14$ changed pixels. This is a very small backdoor compared to an input image of 224×224 pixels for ImageNet models. In other words, the model is very vulnerable, even though it is not poisoned by dirty data. In fact, over 90% of the samples

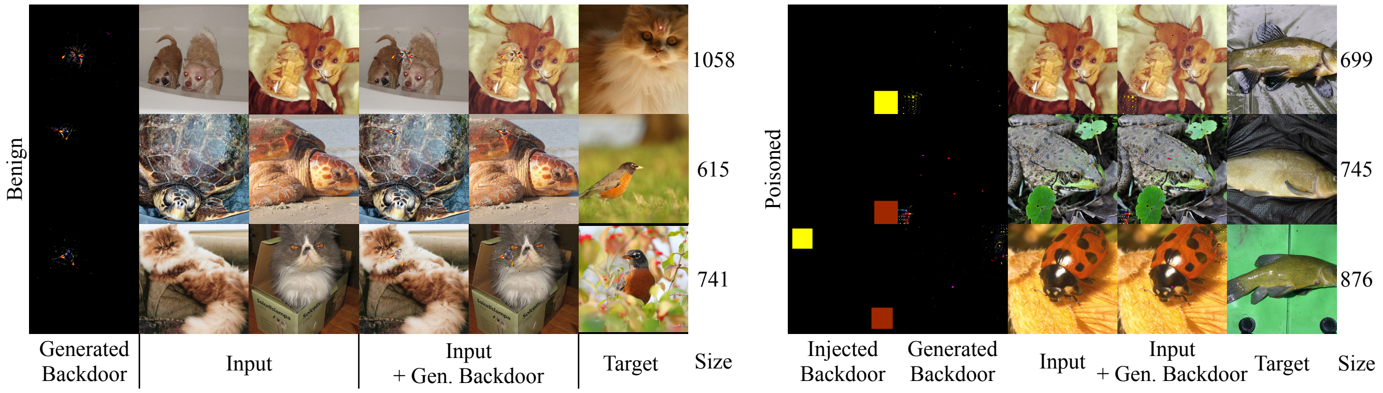

Fig. 1: Natural backdoor versus injected backdoor. The left figure shows natural backdoors (1st column) generated by a model scanner (i.e., NC) for a benign ImageNet model. The right figure presents three poisoned models with injected backdoors with different colors/shapes (1st column). Columns Size denote the size of generated backdoors.

in the victim classes are misclassified to the target class when they are stamped with the small backdoors. Also observe that these backdoors seem to be very low level features that do not constitute any meaningful features in humans' eyes. The model might have overfitted on these low level (strong) features of the target class, causing the natural backdoors. We observe that the backdoor sizes do not meet our intuitions of the distances of these classes. For example, intuitively cat and dog have more similarity than turtle and bird. However, this is not reflected by the sizes of natural backdoors. In some sense, we can say the model is not "*orthogonal*" (more discussion in Section III).

Similar information is presented for three poisoned ImageNet models from [41]. Each model is poisoned in a way that all samples of the victim classes (the third column) stamped with the backdoors in the first column are misclassified to the target class as shown in fifth column. While the models were poisoned with the backdoors shown in the first column, the models overfit on some low-level features of the backdoor patterns used in poisoning such that after poisoning, the injected backdoors are just the small pixel patterns shown in the second column. The last column shows the backdoor sizes. Observe that they are not distinguishable from the sizes of natural backdoors. In the TrojAI multi-round competitions for backdoor scanning organized by IARPA [42], regularly trained clean models have a large number of small natural backdoors whose sizes are not distinguishable from the injected ones in the round 2 competition, which is a round dedicated to finding backdoors in image classification models. As a result, many performers suffered from a large number of false positives (i.e., reporting a clean model as poisoned). On the other hand, if a pretrained model used the hardening technique proposed in the paper and had the hardened class distances published as part of the model specification, the substantially enlarged distances would make stealthy poisoning attempts of the hardened model easily detectable (Section V-C).

Existing Technique I: Adversarial Training. Adversarial training is the most widely used model hardening technique. It aims to train a subject model in such a way that samples in a L^p bound of each training input have the same classification result, with the sacrifice of some classification accuracy. It can move the decision boundary to make the model robust. It can enlarge class distances. In fact, since round 3 of the TrojAI competition, the red team (by NIST) responsible for producing the clean and poisoned models for performers to classify uses adversarial training to suppress natural backdoors. In Figure 2, we show some backdoors for an adversarially trained ImageNet model (downloaded from existing work [48]) in the second column. We use the same three class pairs as in Figure 1, with each pair taking two rows. The natural backdoors are in the second column with their sizes presented on the top of the backdoor images in red in the odd rows. The backdoors are also enlarged in the even rows. The classification confidence of a stamped sample (in the third column) is depicted on the sample.

Comparing to the class distances in Figure 1, it is evident that adversarial training can enlarge class distance (e.g, from 1058 to 1598 for the dog→cat pair). In addition, the backdoors start to possess some human perceptible features. For instance, the backdoor for dog→cat resembles a cat face and the backdoor for turtle→bird has the beak of bird. However, due to the nature of adversarial training, the accuracy has nontrivial degradation (see Section V). Moreover, the order of class distances still does not align well with our intuitions of the two class pairs dog→cat and turtle→bird, indicating that it may not have achieved the maximum class distances. More discussion of the reasons of such insufficiency can be found in Section III.

Existing Technique II: Universal Adversarial Perturbation. While adversarial training generates adversarial perturbations separately for each sample, *universal adversarial perturbation* (UAP) aims to derive common adversarial perturbations for a (large) set of samples from different classes such that the derived UAPs can cause misclassification when they are applied to any samples. Similar to adversarial training, UAPs are usually derived using L^1 . As such, a straightforward idea is to use UAP to adversarially train a model to harden it like in [43]. Figure 3 shows the results. Observe that the derived UAPs in column 1 have noise-like pixel patterns. UAPs can enlarge class distances (e.g, from 1058 to 1118 for the dog→cat pair) to some extent. However, the distance

Input | Backdoor for Adv Model | Backdoor for Our Model | Target

Fig. 2: Adversarial training vs. our training. The 2nd and 4th columns present the generated backdoors for an adversarially trained model and our model, respectively. The patterns highlighted in red boxes are enlarged in even rows. The sizes of backdoors are marked on the top in red. The numbers in 3rd and 5th columns are prediction confidences to the target labels.

1118

872

767

UAP (!10) | Input | Input + UAP | Target Size

Fig. 3: Universal adversarial perturbation. The 1st column shows the generated UAPs for three different victim classes in the 2nd and 3rd columns. The last column presents the sizes of backdoors generated for a hardened model by UAP training.

enlargement is small. This is because UAPs are untargeted, meaning that a UAP for a victim class may flip samples of the class to different target classes. Also note that due to its use of L^1 bound, targeted UAPs are very difficult to derive. They either cannot be found within the bound or the enlarged bound leads to substantial accuracy degradation after hardening. Natural backdoors, on the other hand, do not have any constraints on the magnitude of perturbations. They can easily achieve a high ASR for a specific target, which can reveal vulnerabilities of the decision making between two classes of a subject model.

Technique III: Directly Using Generated Backdoors in Training. Another method is to directly use optimization to derive the smallest backdoor between a pair of classes, e.g., by adapting optimizations in NC [40] and ABS [41]. The training is enhanced such that any samples stamped with the backdoor should retain their ground truth classifications. As we will show in Section V, such a method is expensive as its complexity is quadratic. In addition, its training loss fluctuates a lot (see Section IV-B), causing inferior results in distance improvement compared to ours (with 30% performance difference as shown in Section V-D). Note that the Pairwise baseline evaluated in Section V is already more sophisticated than simply using generated backdoors in hardening as described above. It leverages symmetric hardening and speedup methods discussed later in this section and in detail in Section IV.

Our solution. We propose a novel training method that can effectively enlarge class distances without sacrificing much accuracy. We call it *model orthogonalization* (see Section III). Different from adversarial training that independently perturbs individual inputs, our training considers individual pairs of classes. Specifically, for a pair such as a and b, it derives a minimal backdoor from a to b and another backdoor from b to a. Intuitively, the former can be considered the distinctive (low level) feature of b with respective to a, and vice versa. It then stamps samples with these backdoors and ensures the classification results do not change, which essentially expels these low level features and forces the model to learn high level and more semantic features distinguishing the pair. The two directions of a pair are hardened together, which we call the *symmetric hardening*. As will be discussed in Section IV-B, asymmetric hardening, i.e., hardening only one side or hardening the two sides in separate batches, leads to inferior results. Considering each class pair uniformly (with a total of $\mathcal{O}(n^2)$ pairs for an n-class model) is not cost-effective as different class pairs have different distance capacities. Some pairs quickly reach their maximum distance (e.g., cat and dog) and others need more training to get there. We hence develop a scheduler that schedules the most promising pairs for each batch, substantially improving cost-effectiveness. The training also features a few special designs such as gradually growing optimization bounds and reusing backdoors to speedup the process. Details are in Section IV. The fourth and fifth columns of Figure 2 present example results using our training (on the adversarially trained model). The class distances are substantially enlarged without further accuracy degradation. The distances are much larger compared to the regularly trained model in Figure 1. With hardened distances, the data poisoning in Figure 1 can be easily detected, as the compromised classes would have much smaller distances than the hardened ones. Further, we observe that the order of the distances of the three pairs aligns well with our intuition after hardening. The zoomed-in trigger patterns in the even rows clearly exhibit high level semantic features of the target classes (e.g., the whole body of a bird). In other words, the model learns high level features of individual classes, which substantially mitigates its vulnerabilities to (natural) backdoors.